



KUALA LUMPUR UNIVERSITY
OF SCIENCE AND TECHNOLOGY
(formerly known as Infrastructure University Kuala Lumpur)

Course: UGCS 1001 – Introduction to
Sustainable Infrastructure and Technology

Intelligent Systems in Smart Cities: Building the Future, Today

Group Members:

1. Mohamed Mohamed Hussein – 251925403
2. Hussein Musab Abdi – 251925406
3. Rafi MD Rayhan – 252925890
4. Farah Bashir Mohamed – 252925814
5. Hassan MD Shakib – 252925911



Presentation Outline:

1. Definition and Core Concepts
2. Current Applications in Smart Cities
3. Case Study: Singapore's Smart Traffic Management
4. Enabling Technologies
5. The Benefits of Intelligent Systems
6. The Challenges and Ethical Considerations
7. Future Outlook and Evolution



1. Definition and Core Concepts

Defining Intelligent Systems in a Smart City

A holistic, integrated system that extends beyond AI.



Data Collection: Utilizes technology to gather real-time urban data (e.g., traffic flow, energy consumption).



Data Analysis: Processes this data to derive insights and understand complex urban dynamics.



Informed Decision-Making: Facilitates automated actions to optimize resources and improve the quality of urban life.



Core Functionality: To sense, analyze, and act—functioning as the city's operational brain.

2. Current Applications

Real-World Examples of Intelligent Systems



- **Intelligent Traffic Management:** Dynamically adjusts signal timings to mitigate congestion.
- **Automated Waste Management:** Optimizes collection routes based on sensor data from bins.
- **Smart Public Safety:** Uses analytics to predict and respond to incidents more effectively.
- **Adaptive Street Lighting:** Adjusts illumination based on real-time presence, saving energy.



A Deep Dive into a Real-World Application



The System:

Singapore's "Green Link Determining System" (GLIDE) uses real-time traffic data from vehicle-detection sensors.



How it Works:

The system's AI analyzes traffic volume and flow, then automatically adjusts the green-light time at intersections across the city.



The Technology:

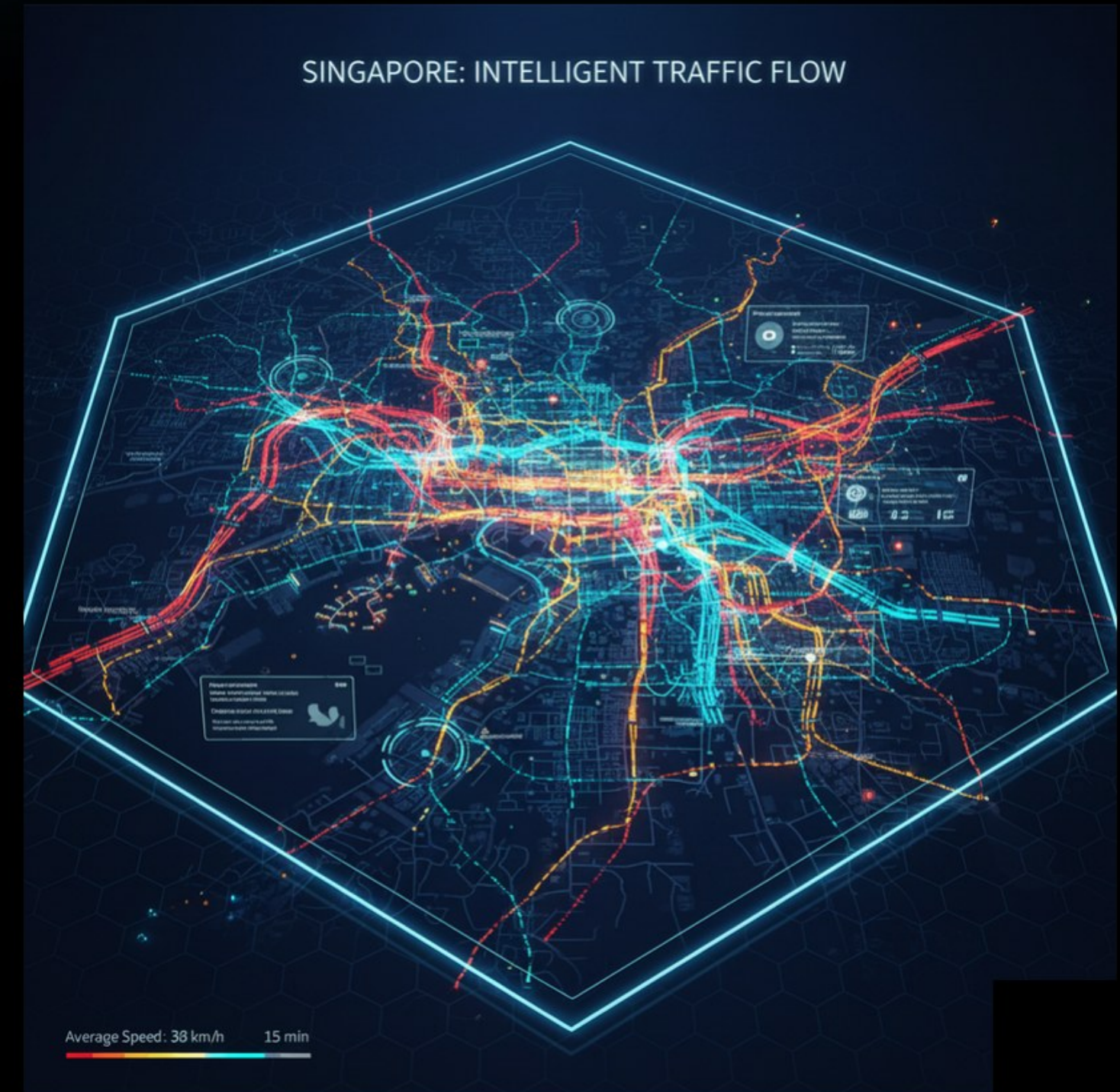
It integrates thousands of sensors, a central control system, and machine learning algorithms that learn traffic patterns.



The Impact:

This intelligent system has successfully reduced average travel times and vehicle congestion, leading to less fuel consumption and lower carbon emissions.

3. Case Study: Singapore's Smart Traffic Management



4. Enabling Technologies



The Key Technologies Behind Smart Cities

- **Internet of Things (IoT):** A vast network of interconnected sensors and devices that serve as the city's data collection backbone.
- **Artificial Intelligence (AI) & Machine Learning:** The analytical core that processes data, identifies patterns, and enables predictive decision-making.
- **Big Data Analytics:** The platform for processing and interpreting the immense volume of data generated by the urban environment.
- **5G Connectivity:** High-speed, low-latency communication infrastructure essential for real-time data transmission between all system components.

5. The Benefits of Intelligent Systems

Analyzing the Advantages for Urban Environments



Enhanced Sustainability: Optimizes energy consumption, reduces waste, and lowers the urban environmental footprint.



Improved Quality of Life: Reduces congestion, shortens commutes, and creates safer, more responsive urban spaces.



Increased Economic Efficiency: Streamlines municipal services, lowers operational costs, and boosts productivity.



Data-Driven Urban Planning: Empowers administrators with actionable insights for smarter, more effective future development.

6. The Challenges and Ethical Considerations

Navigating the Obstacles to Implementation



High Initial Investment: Significant capital is required for the implementation and ongoing maintenance of the technology.



Data Privacy and Governance: Ensuring the ethical collection and secure handling of sensitive citizen data is a critical concern.



Cybersecurity Vulnerabilities: Protecting critical urban infrastructure from sophisticated digital threats is paramount.



Ensuring Digital Equity: We must bridge the technological gap to ensure all citizens can access and benefit from smart city initiatives.

7. Future Outlook

The Evolution of Intelligent Systems in the Next Decade



- **Hyper-Personalization of Services:** Urban services will adapt to individual citizen needs in real-time.
- **Integration of Autonomous Systems:** Expect seamless communication between autonomous vehicles and city infrastructure.
- **Predictive Analytics for Maintenance:** A shift from reactive repairs to proactive maintenance of city infrastructure.
- **Enhanced Citizen Engagement Platforms:** Greater use of digital tools to foster collaboration between citizens and city administration.



Conclusion

Summary of Key Points

- An intelligent system in a smart city is an integrated framework that senses, analyzes, and acts to improve urban operations.
- Proven applications, such as Singapore's traffic system, demonstrate tangible benefits.
- It offers profound benefits but poses significant challenges in cost, privacy, and security that must be carefully managed.
- The system is powered by a convergence of IoT, AI, Big Data, and 5G.
- The system is powered by a convergence of IoT, AI, Big Data, and 5G.



KUALA LUMPUR UNIVERSITY
OF SCIENCE AND TECHNOLOGY
(formerly known as Infrastructure University Kuala Lumpur)

Course: UGCS 1001 – Introduction to
Sustainable Infrastructure and Technology

Thank You for Your Attention



Lecturer: Assoc. Prof. Ts. Sr Gs. Ranjit Singh

